

Seungmin Oh 오승민

Ph.D. candidate in Artificial Intelligence

Multi-Modal Artificial Intelligence Lab, Ajou University, South Korea

Advised by Prof. Jongbin Ryu, Ph.D. expected Feb 2028

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Suwon, South Korea 📍

Ph.D. candidate building **efficient AI** for deployment under tight compute limits, from edge devices to robot control loops, across vision, LLM, VLM, and VLA models.

Education

Integrated M.S./Ph.D. in Artificial Intelligence

Ajou University, Suwon, South Korea, advised by Prof. Jongbin Ryu

Feb 2022 - present

B.S. in Software and Computer Engineering

Ajou University, Suwon, South Korea

Feb 2016 - Feb 2022

Research experience

Multi-Modal Artificial Intelligence Lab, Ajou University

Graduate Researcher, advised by Prof. Jongbin Ryu

Feb 2022 - present

Vision-language-action models for robotic manipulation

2025 - present

- Training methodology: world-state change prediction instead of raw actions, preserving VLM grounding (in prep.).
- Inference acceleration: fast action controller with lazy gating; full VLA invoked only on anticipated failure (in prep.).
- Failure detection: catching execution failures early enough to recover [S.1].

Efficient LLM compression

2024 - present

- Structured pruning: overcomplete training for scaled recovery capacity, up to +8.4 points at zero deployment cost [C.3].
- Structured pruning: curriculum-based layer-wise compression schedule, 1.7× faster convergence at half the GPU memory [C.4].

Efficient vision and test-time adaptation

2022 - present

- Test-time adaptation: candidate verification by divergence shift, up to 2× faster with over 3× less memory [C.5].
- On-device classification: 1st place, 2025 IEEE Low-Power Computer Vision Challenge Track 1 [A.1].
- Re-parameterization: branch-level neural substitution [C.1].
- Representation learning: transformation-aware prompt conditioning via re-calibrated contrastive loss [C.2].
- Medical: CT-based triage for small-bowel obstruction [J.1]; CNN vs. ViT architectural behavior [J.2].

Funded applied research projects (ETRI, NIA, 42dot)

2021 - 2024

- Real-time detection transformer for autonomous driving (42dot); source-free domain adaptation for on-device face recognition (ETRI); lymphoma recognition and survival analysis from 3D CT (NIA); masked-face recognition with domain adaptation (ETRI).

GGQ Company Inc., Seoul, South Korea

Research Intern

2020

- Game-scene understanding: player-name, skill, and HUD OCR in League of Legends; face recognition for attendance.

Publications

S under review, C conference, A preprint, J journal, P patent

- [S.2] Yuri Kim, **Seungmin Oh**, Jongbin Ryu. Adaptive regularization. *Under review*.
- [S.1] Donghyun Kim, **Seungmin Oh**, Dooyeon Na, Jongbin Ryu. Failure detection for VLA models. *Under review*.
- [C.5] **Seungmin Oh**, Seunghun Kang, Jongbin Ryu. Efficient Test-Time Adaptation through Candidate Verification and Divergence Shifts. *NeurIPS*, 2026.
- [C.4] Donggeon Lee, Dooyeon Na, **Seungmin Oh**, Jongbin Ryu. Layer-wise Curriculum Learning for Efficient LLM Compression. *EMNLP*, 2026.
- [C.3] **Seungmin Oh**, Donggeon Lee, Jongbin Ryu. Train Overcomplete, Deploy Compact: Scaling Recovery Capacity for Structured LLM Pruning. *EMNLP*, 2026.
- [C.2] **Seungmin Oh**, Seunghun Kang, Jongbin Ryu. Re-calibrated Contrastive Loss for Transformation-Aware Prompt Conditioning in Vision-Language Models. *BMVC*, 2026.
- [A.1] Zihao Ye et al., including **Seungmin Oh**, Jongbin Ryu. Evaluation of Winning Solutions of 2025 Low Power Computer Vision Challenge. *arXiv preprint*, 2025.
- [C.1] **Seungmin Oh**, Jongbin Ryu. Neural Substitution for Branch-Level Network Re-parameterization. *ACCV*, 2024.
- [J.2] **Seungmin Oh**, Namkug Kim, Jongbin Ryu. Analyzing to Discover Origins of CNNs and ViT Architectures in Medical Images. *Scientific Reports*, 14:8755, 2024.

- [J.1]

Seungmin Oh, Jongbin Ryu, Hojung Shin, Jeongho Song, Sangyong Son, Hoon Hur, Sanguk Han. Deep Learning Using Computed Tomography to Identify High-Risk Patients for Acute Small Bowel Obstruction: Development and Validation of a Prediction Model. [International Journal of Surgery](#), 109(12), 2023 (IF 15.3, JCR top 1%).
- [P.2]

Jongbin Ryu, Dooyeon Na, Seungmin Oh, Donggeon Lee. System and Method for Compressing a Deep Learning Model Based on Layer-Wise Curriculum Learning. *Korean Patent Application No. 10-2026-0123632*, filed Jul 2026.
- [P.1]

Hojung Shin, Jongbin Ryu, Seungmin Oh. Electronic Device and Method for Classifying Small Bowel Obstruction Using Abdominal Computed Tomography Images. *PCT International Patent Application No. PCT/KR2024/014604*, filed Sep 2024.

Honors and awards

IEEE Low-Power Computer Vision Challenge, Winner CVPR 2025 Workshops, organized by IEEE, CVF, Qualcomm, and Purdue University; ranked 1st among 45 teams Layer Imitation: Efficient Vision-Language Model on the Mobile NPU ↗	Apr 2025
Google Solution Challenge, Global Top 100 Google 'Smiler,' an expression-training app for children with ASD ↗	May 2023
ICT Mentoring Conference, Excellence Award ACK, KOSIA Recommender System for Web Search Based on NLP to Improve User Search Environment ↗	Dec 2021
Student Paper Competition, First Prize Korea Digital Contents Society Lightweight Video Search Using Video-to-Text Generation and Retrieval	Jul 2021

Research funding

Doctoral Research Grant National Research Foundation of Korea (NRF), \$36K Progressive Linear Activation for Depth-wise Reparameterization of Large-scale Models	Sep 2025 - Aug 2027
Lesion-Class Re-labeling in Medical Image Training Data National Information Society Agency (NIA), \$20K	Oct 2022 - Dec 2023

Technical skills

Infrastructure	Built and operate a multi-node GPU cluster (about 20 nodes, 8× RTX 3090 each): containerized environment provisioning, per-node CUDA isolation, unattended long-running training pipelines.
Programming	Python; C/C++ with SIMD- and cache-aware performance optimization; CUDA.
ML tooling	PyTorch, Hugging Face, Weights & Biases, DVC, Docker, Kubernetes, Git and GitHub Actions.
On-device	Qualcomm AI Hub (Snapdragon NPU); structured pruning and re-parameterization for compact deployment.

Teaching and leadership

Lecturer, tutorials for undergraduate AI research assistants, Ajou University (3 terms)	2023 - 2024
Teaching Assistant, Capstone Design Project, Ajou University	2023
Founding Member, Google Developer Student Club, Ajou University chapter	Sep 2022 - Dec 2023